

- Vertical + Horizontal shifts: p.37
- Vertical + Horizontal Stretching/Shrinking: p.37.
- Combinations of functions: p.40.
- Composite functions: p.41.

$$(f \circ g)(x) = f(g(x))$$

$$\rightarrow D_{f \circ g} = \{x \mid x \in D_g \text{ and } g(x) \in D_f\}$$

Examples:

1) $f(x) = \sqrt{x-8}$, $g(x) = x^2$. Find

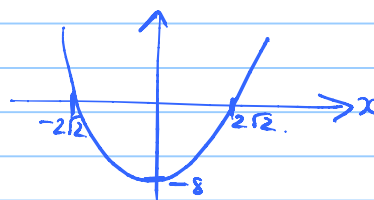
(a) $(f \circ g)(x)$ and $D_{f \circ g}$.

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= f(x^2) \\ &= \sqrt{x^2 - 8} \end{aligned}$$

$$D_{f \circ g} = \{x \mid x^2 - 8 \geq 0\}$$

$$= (-\infty, -2\sqrt{2}] \cup [2\sqrt{2}, \infty).$$

We use a graph to solve
 $x^2 - 8 \geq 0$



(b) $(g \circ f)(x)$ and $D_{g \circ f}$

$$\begin{aligned} (g \circ f)(x) &= g(f(x)) \\ &= g(\sqrt{x-8}) \\ &= (\sqrt{x-8})^2 \\ &= x-8. \end{aligned}$$

$$\begin{aligned} x-8 &\geq 0 \\ \Rightarrow x &\geq 8 \\ \Rightarrow D_{g \circ f} &= [8, \infty) \end{aligned}$$

2) Find $f(x)$ and $g(x)$ such that
 $F(x) = (f \circ g)(x)$, where $F(x) = \frac{1}{x-3}$

$$F(x) = f(g(x)) = \frac{1}{x-3}.$$

Let $g(x) = x-3$ and $f(x) = \frac{1}{x}$.

③ Find f, g and h such that $F = f \circ g \circ h$, $F(x) = 1 - 3^{x^2}$

$$F(x) = 1 - 3^{x^2} = f(g(h(x)))$$

$$= f(g(x^2)) \quad \text{if } h(x) = x^2$$

$$= f(3^{x^2}) \quad \text{if } g(x) = 3^x$$

$$= 1 - 3^{x^2} \quad \text{if } f(x) = 1 - x.$$

$\Rightarrow$ We choose
 $f(x) = 1 - x$
 $g(x) = 3^x$
 and $h(x) = x^2$.